

Machine Learning
Program Elective -- UG3
L-T-P-C: 2-1-0-3

1. Course Objectives:

This course provides the statistical inference and underlying mathematical relationships across various machine learning algorithms. This course also provides exposure to design, implement and analyse the different machine learning algorithms for a given problem.

2. Syllabus:

Unit – 1 : Introduction to machine learning (ML): Supervised and Unsupervised Learning - Regression, Classification and Clustering. Performance metrics of ML models for classification and regression.

Unit – 2 : Regression – gradient descent optimization, regularized least squares (Ridge, Lasso etc.). Classification – Logistic Regression, Decision Trees. Bias-Variance trade-off..

Unit – 3 : Probabilistic models – Bayesian Classifier, Naive Bayes Classifier-Text Classification. Non-parametric techniques – Parzen Windows, K-nearest neighbor classification.

Unit – 4 : Neural Networks, Simple Perceptron - Linearly separable case. Multilayer Perceptron, Activation Functions, Backpropagation algorithm. Performance tuning – Cross Validation.

Unit – 5 : Support vector machines – hard, soft, and nonlinear problem formulations, Kernels, and KKT conditions.

Unit – 6 : Unsupervised Learning: Introduction to clustering and dimensionality reduction techniques – PCA and SVD, k-means clustering and Gaussian Mixture Model (GMM) clustering.

3. Course Outcomes:

At the end of the course, students should have the ability:

- i) Able to explain the fundamental concepts of machine learning including data and model complexity.
- ii) To choose or design an appropriate machine learning algorithm for solving a given problem.
- iii) To perform the dimensionality reduction to deal with the dynamically growing data.
- iv) To analyze the performance of given or implemented machine-learning solutions on practical datasets.

4. Text Books:

- a) Bishop, Christopher M., Pattern Recognition and Machine Learning, Springer, 2006, ISBN 978-0-387-31073-2
- b) Tom Mitchell, Machine Learning, McGraw Hill, 1997, ISBN 0070428077

5. Reference Books:

- a) Ethem Alpaydin, Introduction to Machine Learning, Fourth Edition, The MIT Press,

October 2004, ISBN 0-262-01211-1

b) Richard O. Duda, Peter E. Hart, David G. Stork, Pattern Classification, 2nd Edition, Publisher John Wiley & Sons, 2006, ISBN: 978-0-471-05669

c) M. Narasimha Murty, V. Susheela Devi, Pattern Recognition An Algorithmic Approach, Springer, 2011, ISBN 978-0-85729-495-1

d) James, G., Witten, D., Hastie, T., Tibshirani, R, An Introduction to Statistical Learning, Springer, 2013, 978-1-4614-7137-0

5.Evaluation:

Quiz: 10%

Mid: 25%

Term Paper: 15%

Endsem: 50%